"Empowerment through quality technical education"

Ajeenkya DY Patil School of Engineering

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Department of Computer Engineering

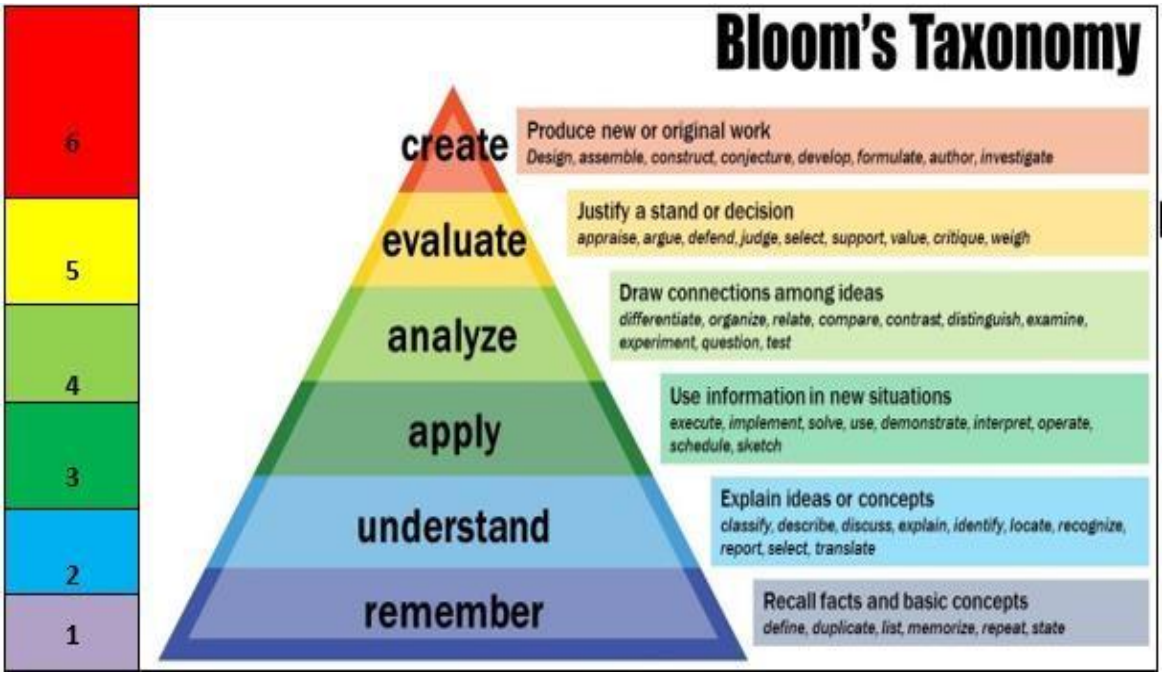
Assignments **Form No. IQAC/18a1**

Class: SE Div: C	Course Name: Data Structures	Course Coordinator: Prof.Ishwar Bharambe
Assignment No.3	Assignment based on Units: 3	Date of issue: 18/09/2026

CO 3: Make Use of Stacks and Queues to solve the given problem

Sr. No.	Questions	Level of Mapping and Number			Blooms Level
		CO	PO	PSO	
Q.1.	Explain different stack operations with suitable algorithms and write a program to implement multiple stacks in a single array.	CO3	PO 1, PO5	PSO 1	1
Q.2.	Convert the following infix expression into prefix and postfix expressions. Show each step clearly. Expression: [(A + B) * (C - D) / E]	CO3	PO2,PO3	PSO 2	2
Q.3.	Design an algorithm to evaluate a postfix expression. Implement the algorithm in any programming language and trace it for the input: 6 3 2 + * 4 -	CO3	PO1, PO 4	PSO 1-2	3
Q.4.	Implement circular queue using an array. Explain how it overcomes the limitation of a simple queue. Discuss at least two real-world applications of circular queue.	CO3	PO 1, PO 6	PSO 1	4
Q.5.	Compare singly linked list, doubly linked list, and circular linked list in terms of memory usage, traversal efficiency, and applications.	CO3	PO 2-3, PO11	PSO3	2

Sign of Course Coordinator: Prof.Ishwar Bharambe	Sign of HoD


Bloom's Taxonomy
6
5
4
3
2
1
create
Produce new or original work
Design, assemble, construct, conjecture, develop, formulate, author, investigate
evaluate
Justify a stand or decision
appraise, argue, defend, judge, select, support, value, critique, weigh
analyze
Draw connections among ideas
differentiate, organize, relate, compare, contrast, distinguish, examine, experiment, question, test
apply
Use information in new situations
execute, implement, solve, use, demonstrate, interpret, operate, schedule, sketch
understand
Explain ideas or concepts
classify, describe, discuss, explain, identify, locate, recognize, report, select, translate
remember
Recall facts and basic concepts
define, duplicate, list, memorize, repeat, state